विशाल जाधव सरांचे

VJTech Academy

Inspiring Your Success...

“UNIT-II Classes & Objects”

Program: Classes & Objects

```
class Addition
{
	int a,b,c;
	void getdata()
	{
		a=100;
		b=50;
	}
	void putdata()
	{
		c=a+b;
		System.out.println("Addition="+c);
	}
	public static void main(String args[])
	{
		Addition a1=new Addition();
		a1.getdata();
		a1.putdata();
	}
}
```

Program: WAP to declare Student having data members rollno, name and marks. Accept and display data for one student.

```
import java.lang.*;
import java.util.*;
class Student
{
	int rollno;
	String name;
	float marks;
	void get_stud_info()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Student Roll NO:");
		rollno=sc.nextInt();
		System.out.println("Enter Student Name:");
		name=sc.next();
		System.out.println("Enter Student Marks:");
		marks=sc.nextFloat();
	}
	void disp_stud_info()
	{
		System.out.println("Roll No:"+rollno);
		System.out.println("Name:"+name);
		System.out.println("Marks:"+marks);
	}
	public static void main(String args[])
	{
		Student s1=new Student();
		s1.get_stud_info();
		s1.disp_stud_info();
	}
}
```

Program: to declare Book having data members bookid, name and price. Accept and display data for two objects.

```
import java.util.*;
class Book
{
	int bookid;
	String name;
	float price;
	void get_book_info()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Book ID:");
		bookid=sc.nextInt();
		System.out.println("Enter Book Name:");
		name=sc.next();
		System.out.println("Enter Book Price:");
		price=sc.nextFloat();
	}
	void disp_book_info()
	{
		System.out.println("Book ID:"+bookid);
		System.out.println("Book Name:"+name);
		System.out.println("Book Price:"+price);
	}
	public static void main(String args[])
	{
		Book b1=new Book();
		Book b2=new Book();
		b1.get_book_info();
		b2.get_book_info();
		b1.disp_book_info();
		b2.disp_book_info();
	}
}
```

Program: Array of objects

```
import java.util.*;
class Employee
{
	int empid;
	String name;
	float salary;
	void get_emp_info()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Employee ID:");
		empid=sc.nextInt();
		System.out.println("Enter Employee NAME:");
		name=sc.next();
		System.out.println("Enter Employee Salary:");
		salary=sc.nextFloat();
	}
	void disp_emp_info()
	{
		System.out.println(empid+"\t"+name+"\t"+salary);
	}
	public static void main(String argsp[])
	{
		Employee e[]=new Employee[5];
		int i;
		for(i=0;i<5;i++)
		{
			e[i]=new Employee();
			e[i].get_emp_info();
		}
		System.out.println("EMPID\tNAME\tSALARY");
		System.out.println("====================");
		for(i=0;i<5;i++)
		{
			e[i].disp_emp_info();
		}
	}
}
```

Program: Array of objects (two classes in single java file)

```
import java.util.*;
class Country
{
	int country_code;
	String country_name;
	void get_country_info()
	{
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter Country Code:");
		country_code=sc.nextInt();
		System.out.println("Enter Country NAME:");
		country_name=sc.next();
	}
	void disp_country_info()
	{
		System.out.println(country_code+"\t"+country_name);
	}
}
class MainProgram
{
	public static void main(String args[])
	{
		//datatype arrayname[]=new datatype[SIZE];
		Country e[]=new Country[5];
		int i;
		for(i=0;i<5;i++)
		{
			e[i]=new Country();
			e[i].get_country_info();
		}
		System.out.println("CountryCode\tCountryName");
		System.out.println("==========================");
		for(i=0;i<5;i++)
		{
			e[i].disp_country_info();
		}
	}
}
```

Program: Default Constructor

```
class DefaultConstructor
{
	int a,b,c;
	DefaultConstructor()
	{
		a=100;
		b=50;
	}
	void display()
	{
		c=a+b;
		System.out.println("Addition="+c);
	}
	public static void main(String args[])
	{
		DefaultConstructor d1=new DefaultConstructor();
		d1.display();
	}
}
```

Program: Parameterized Constructor

```
class ParameterizedConstructor
{
	int a,b,c;
	ParameterizedConstructor(int x,int y)
	{
		a=x;
		b=y;
	}
	void display()
	{
		c=a+b;
		System.out.println("Addition="+c);
	}
	public static void main(String args[])
	{
		ParameterizedConstructor p1=new ParameterizedConstructor(10,20);
		p1.display();
	}
}
```

Program: Copy Constructor

```
class Sample
{
	int a,b,c;
	Sample()
	{
		a=10;
		b=20;
	}
	Sample(Sample m)
	{
		a=m.a;
		b=m.b;
	}
	void display()
	{
		c=a+b;
		System.out.println("Addition="+c);
	}
	public static void main(String args[])
	{
		Sample c1=new Sample();
		Sample c2=new Sample(c1);
		System.out.println("Object c1");
		c1.display();
		System.out.println("Object c2");
		c2.display();
	}
}
```

Program: Constructor Overloading

```
class Sample1
{
	int a,b,c;
	Sample1()
	{
		a=10;
		b=20;
	}
	Sample1(int x)
	{
		a=x;
		b=200;
	}
	Sample1(int m,int n)
	{
		a=m;
		b=n;
	}
	void display()
	{
		c=a+b;
		System.out.println("Addition="+c);
	}
	public static void main(String args[])
	{
		Sample1 c1=new Sample1();
		Sample1 c2=new Sample1(100);
		Sample1 c3=new Sample1(1000,2000);
		System.out.println("Object c1");
		c1.display();
		System.out.println("Object c2");
		c2.display();
		System.out.println("Object c3");
		c3.display();
	}
}
```

Program: static data member and static member function

```
class StaticDemo
{
		int no;
		static int count;
		void getdata(int x)
		{
			no=x;
			count++;
		}
		void disp_no()
		{
			System.out.println("Value of no="+no);
		}
		static void disp_count()
		{
			System.out.println("Value of count="+count);
		}
		public static void main(String args[])
		{
			StaticDemo s1=new StaticDemo();
			StaticDemo s2=new StaticDemo();
			StaticDemo s3=new StaticDemo();
			s1.getdata(100);
			s2.getdata(200);
			s3.getdata(300);
			System.out.println("***Object s1***");
			s1.disp_no();
			StaticDemo.disp_count();
			System.out.println("***Object s2***");
			s2.disp_no();
			StaticDemo.disp_count();
			System.out.println("***Object s3***");
			s3.disp_no();
			StaticDemo.disp_count();
		}
}
```

Program: Single Dimensional Array

```
class OneDArray
{
			public static void main(String args[])
			{
				//int a[]={10,20,30,40,50};
				int a[]=new int[5];
				a[0]=10;
				a[1]=20;
				a[2]=30;
				a[3]=40;
				a[4]=50;

				System.out.println("Array Elements:");
				for(int i=0;i<a.length;i++)
				{
					System.out.println(a[i]);
				}
			}
}
```

Single Dimensional Array with Scanner class

```
import java.util.*;
class OneDArray1
{
	public static void main(String args[])
	{
		int a[]=new int[5];
		int i;
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter five array elements:");
		for(i=0;i<5;i++)
		{
			a[i]=sc.nextInt();
		}
		System.out.println("Array Elements:");
		for(i=0;i<a.length;i++)
		{
			System.out.println(a[i]);
		}
	}
}
```

One dimensional Array : reverse the elements of array

```
import java.util.*;
class OneDArray2
{
	public static void main(String args[])
	{
		int a[]=new int[5];
		int i;
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter five array elements:");
		for(i=0;i<a.length;i++)
		{
			a[i]=sc.nextInt();
		}
		System.out.println("Array Elements in reverse order:");
		for(i=a.length-1;i>=0;i--)
		{
			System.out.println(a[i]);
		}
	}
}
```

//Two Dimensional Array

```
class TwoDArray
{
	public static void main(String args[])
	{
		/*int a[][]={{10,20,30},
					{40,50,60},
					{70,80,90}
				   }; */
		int a[][]=new int[3][3];
		a[0][0]=10;
		a[0][1]=20;
		a[0][2]=30;
		a[1][0]=40;
		a[1][1]=50;
		a[1][2]=60;
		a[2][0]=70;
		a[2][1]=80;
		a[2][2]=90;

		System.out.println("Element present at 0 row 0 column ="+a[0][0]);
		System.out.println("Element present at 0 row 1 column ="+a[0][1]);
		System.out.println("Element present at 0 row 2 column ="+a[0][2]);
		System.out.println("Element present at 1 row 0 column ="+a[1][0]);
		System.out.println("Element present at 1 row 1 column ="+a[1][1]);
		System.out.println("Element present at 1 row 2 column ="+a[1][2]);
		System.out.println("Element present at 2 row 0 column ="+a[2][0]);
		System.out.println("Element present at 2 row 1 column ="+a[2][1]);
		System.out.println("Element present at 2 row 2 column ="+a[2][2]);

	}
}
```

```
Element present at 0 row 0 column =10
Element present at 0 row 1 column =20
Element present at 0 row 2 column =30
Element present at 1 row 0 column =40
Element present at 1 row 1 column =50
Element present at 1 row 2 column =60
Element present at 2 row 0 column =70
Element present at 2 row 1 column =80
Element present at 2 row 2 column =90
```

//Two Dimensional Array

```
import java.util.*;
class TwoDArray1
{
	public static void main(String args[])
	{
		int a[][]=new int[3][3];
		int i,j;
		Scanner sc=new Scanner(System.in);
		System.out.println("Enter 3*3 Matrix Elements:");
		for(i=0;i<3;i++)
		{
			for(j=0;j<3;j++)
			{
				a[i][j]=sc.nextInt();
			}
		}
		System.out.println("Your 3*3 Matrix Elements:");
		for(i=0;i<3;i++)
		{
			System.out.print("\n");
			for(j=0;j<3;j++)
			{
				System.out.print(a[i][j]+"\t");
			}
		}
	}
}
```

```
Enter 3*3 Matrix Elements:
11
22
33
44
55
66
77
88
99
Your 3*3 Matrix Elements:

11	22	33
44	55	66
77	88	99
```

/*Write a program to create vector with six elements (10,30,60,70,80,100). Removes element 3rd and 4th position. Insert new element at 3rd position. Display the original and current size of vector */

```
import java.util.*;
class VectorDemo1
{
	public static void main(String args[])
	{
		Vector v1=new Vector();

		v1.addElement(new Integer(10));
		v1.addElement(new Integer(30));
		v1.addElement(new Integer(60));
		v1.addElement(new Integer(70));
		v1.addElement(new Integer(80));
		v1.addElement(new Integer(100));
		System.out.println("Initial Vector Size = "+v1.size());
		v1.removeElementAt(3);
		v1.removeElementAt(4);
		v1.insertElementAt(new Integer(150),3);
		System.out.println("Final Vector Size = "+v1.size());
	}
}
```

Vector class is collection of objects and value taken by using command line arguments

```
import java.util.*;
class VectorDemo2
{
	public static void main(String args[])
	{
		Vector v1=new Vector();
		for(int i=0;i<args.length;i++)
		{
			v1.addElement(args[i]);
		}
		System.out.println("Vector of Size = "+v1.size());
		System.out.println("Your Vector Elements:");
		for(int i=0;i<v1.size();i++)
		{
			System.out.println(v1.elementAt(i));
		}
	}
}
```

```
F:\JavaBatch2024\UNIT-II>java VectorDemo2 100 200 300
Vector of Size = 3
Your Vector Elements:
100
200
300
```

//String class

```
class StringDemo
{
	public static void main(String args[])
	{
		String str=new String("VJTech");

		System.out.println("Value of str ="+str);
		System.out.println("Length of str ="+str.length());
		System.out.println("To Lower Case ="+str.toLowerCase());
		System.out.println("To Upper Case ="+str.toUpperCase());
		System.out.println("Character present at 2 index ="+str.charAt(2));
		System.out.println("Concatenation="+str.concat("Academy"));
		System.out.println("Index of e character ="+str.indexOf('e'));
		System.out.println("Equals method ="+str.equals("VJTech"));
		System.out.println("CompareTo method ="+str.compareTo("VJTech"));
	}
}
```

```
Value of str =VJTech
Length of str =6
To Lower Case =vjtech
To Upper Case =VJTECH
Character present at 2 index =T
Concatenation=VJTechAcademy
Index of e character =3
Equals method =true
CompareTo method =0
```

StringBuffer class

```
class StringBufferDemo
{
	public static void main(String args[])
	{
		StringBuffer str=new StringBuffer("VJTech");

		System.out.println("Original String:"+str);
		System.out.println("Length of String:"+str.length());
		for(int i=0;i<str.length();i++)
		{
		  System.out.println("Character at position "+i+" is "+str.charAt(i));
		}
		str.setCharAt(3,'T');
		System.out.println("Modified String :"+str);
		str.append("Academy");
		System.out.println("Appended String :"+str);
	}
}
```

Original String:VJTech
Length of String:6
Character at position 0 is V
Character at position 1 is J
Character at position 2 is T
Character at position 3 is e
Character at position 4 is c
Character at position 5 is h
Modified String :VJTTch
Appended String :VJTTchAcademy

Wrapper class

```
class WrapperClassDemo
{
	public static void main(String args[])
	{
		int i=100;
		Integer ii=new Integer(i);
		System.out.println("Primitive Integer Value = "+i);
		System.out.println("Object Integer Value = "+ii);
	}
}
```